

Honors Geometry - 3.1 / 3.2 Practice

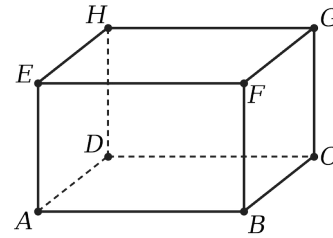
Name: _____

GAME DAY — Pairs of Lines and Angles • Parallel Lines and Transversals

Run every play, find your answer in that drive's answer bank, and write its letter in the box with the play's number. Each drive spells an NFL team. If your drive spells nonsense, a play went wrong — find it before moving on.

DRIVE 1 — Pregame: the press box Use the box for plays 1-4.

- Which line is SKEW to line AB: line DC, line AE, line CG, or line EF?
- Which plane is PARALLEL to plane ABC: plane BCG, plane EFG, or plane ADH?
- Name the line through point D that is parallel to line AE.
- How many lines through point E are parallel to line BC?
Which postulate says so?



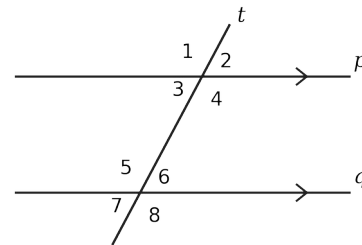
line DC A	exactly one S	line BF K	plane EFG E	line AE O	line DH T
infinitely many P	plane BCG I	line CG J	line EF U	none L	plane ADH R

Drive 1 spells:

#1	#2	#3	#4
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DRIVE 2 — First quarter: read the field Classify each pair of angles.

- $\angle 1$ and $\angle 5$ _____
- $\angle 3$ and $\angle 6$ _____
- $\angle 2$ and $\angle 7$ _____
- $\angle 4$ and $\angle 6$ _____
- $\angle 5$ and $\angle 8$ _____



alternate exterior L	linear pair R	corresponding C	vertical angles S
none of these N	consecutive interior T	alternate interior O	

Drive 2 spells:

#5	#6	#7	#8	#9
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DRIVE 3 — Second quarter: angle measures $p \parallel q$. New measures each play, same numbering.

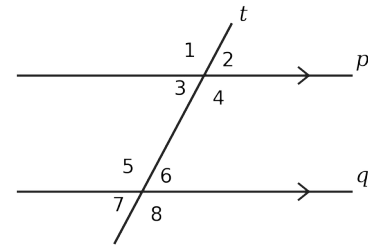
10. $m\angle 1 = 112^\circ$. Find $m\angle 6$.

11. $m\angle 1 = 112^\circ$. Find $m\angle 8$.

12. $m\angle 3 = 57^\circ$. Find $m\angle 6$.

13. $m\angle 4 = 101^\circ$. Find $m\angle 6$.

14. $m\angle 7 = 44^\circ$. Find $m\angle 2$.



Name the theorem you used for each play next to your answer.

68° B	123° G	44° S	101° H	112° E
22° M	79° R	136° K	57° A	

Drive 3 spells:

#10	#11	#12	#13	#14
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DRIVE 4 — Second half: solve for x $p \parallel q$, same diagram. Name the pair, then write the equation.

15. $m\angle 2 = (4x + 10)^\circ$ and $m\angle 6 = 70^\circ$

18. $m\angle 2 = (7x - 6)^\circ$ and $m\angle 7 = (4x + 27)^\circ$

16. $m\angle 3 = (5x - 12)^\circ$ and $m\angle 6 = (3x + 20)^\circ$

19. In play 17, find $m\angle 4$.

17. $m\angle 4 = (4x + 45)^\circ$ and $m\angle 6 = (2x + 15)^\circ$

20. Finish the proof of the Alternate Exterior Angles Theorem. Given: $p \parallel q$. Prove: $\angle 1 \cong \angle 8$

Statements	Reasons
1. $p \parallel q$	1. Given
2. $\angle 1 \cong \angle 5$	2. _____
3. $\angle 5 \cong \angle 8$	3. Vertical Angles Theorem
4. $\angle 1 \cong \angle 8$	4. Transitive Property of Congruence

$x = 25$ D	Corresponding Angles Theorem S	$x = 11$ W	$x = 21.5$ C
$x = 15$ B	Vertical Angles Theorem Q	125° N	$x = -15$ F
$x = 20$ O	55° T	$x = 16$ R	Alternate Interior Angles Theorem Z

Drive 4 spells:

#15	#16	#17	#18	#19	#20
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